1 Planning (15 marks)

Defining the problem (3 marks)

P1 Frequency or period of rotation or ω is the independent variable and θ is the dependent variable or vary f or T or ω and measure θ. [1]

P2 $\omega = 2\pi f = 2\pi/T$ [1]

P3 Keep the length of the rigid rod constant; ignore reference to mass. [1]

Methods of data collection (5 marks)

M1 Labelled diagram of apparatus: small object, pole attached to a rotating device (motor, turntable). [1]

M2 Method to change the speed of the rotating device. [1]

M3 Method to determine frequency or time period (e.g. stop watch to time a number of rotations, rev counter/tachometer, light gates connected to a timer/frequency meter). [1]

M4 Use fiducial mark or light gates perpendicular to motion of object. [1]

M5 Method to measure angle – use protractor or rule for measurements for trigonometry methods. This must be shown correctly on diagram or explained in text. [1]

Method of analysis (2 marks)

A1 Plot a graph of cos θ against $1/\omega^2$. [1]

A2 Relationship is valid if straight line through the origin [1]

Safety considerations (1 mark)

S1 Use a protective screen in case mass detaches from the pole. Do not use goggles. [1]

Additional detail (4 marks)

Relevant points might include [4]

1 Large motor speed to produce measurable θ.
2 Additional detail on measuring angle e.g. large protractor fixed to pole.
3 Projection method, slow motion freeze frame video, camera with detail.
4 $\cos\theta = h/l$ or equivalent.
5 Method of checking pole is vertical – use a set square.
6 Additional detail on measuring angular velocity, e.g. time at least 10 rotations.
7 Wait for motion to become stable.

Do not allow vague computer methods.

[Total: 15]

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